

SOFTWARE REQUIREMENTS SPECIFICATION

Railway Ticket Reservation System

Version: 1.0
Date: 06-09-2026
Status: Draft for Mini Project Submission

Role	Name	USN	Responsibility
Product Owner	Aryaprasad	PES1UG24AM331	Requirements, priorities, acceptance and project direction
Developer	Deeptanshu Kumar	PES1UG24AM357	Implementation and technical development
Tester	Amar	PES1UG24AM340	Test planning, execution, defect reporting and verification

Revision History

Version	Date	Author	Change Summary
1.0	06-09-2026	Project Team	Initial SRS prepared for Railway Ticket Reservation System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Railway Ticket Reservation System. The system is intended to provide a simple digital interface through which passengers can search trains, check seat availability, book tickets, view booking details, and cancel bookings. The document serves as the baseline for development, testing, Jira task creation, and acceptance of the mini project.

1.2 Scope

The system covers passenger registration and authentication, train search, route and schedule display, seat availability checking, ticket booking, booking confirmation, booking history, ticket cancellation, refund-status recording, and basic administrative management of train and schedule information. The project is limited to the application-level reservation workflow; actual railway operational control, train movement, and real-world railway backend systems are outside the scope of this mini project.

1.3 Audience

The intended audience includes the student development team, tester, product owner, course instructor/evaluator, and future maintainers of the project.

1.4 Definitions and Acronyms

Term	Meaning
RTS	Railway Ticket Reservation System
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
RTM	Requirements Traceability Matrix
PNR	Passenger Name Record / booking reference
UI	User Interface
API	Application Programming Interface
Admin	Authorized system administrator
Authentication	Process of verifying a user's identity

2. Overall Description

2.1 Product Perspective

The Railway Ticket Reservation System is a web-based application. Passengers interact with the system through a browser-based user interface. The application communicates with a backend service and a database to authenticate users, retrieve train information, maintain seat availability, create reservations, and process cancellations. An administrator manages train, route, schedule, fare, and availability information through an administrative interface.

2.2 Major Product Functions

- User registration and login.
- Search trains using source, destination and journey date.
- Display matching trains, departure/arrival information and fare.
- Display current seat availability.
- Allow a passenger to select a train and enter passenger details.
- Create and confirm a booking with a unique booking/PNR number.
- Display and store booking history.
- Allow a passenger to view an existing ticket.
- Allow cancellation of an eligible booking.

- Calculate and record the applicable cancellation/refund amount.
- Update seat availability after successful booking or cancellation.
- Provide an administrator interface for train and schedule management.
- Maintain records required for auditing and troubleshooting.

2.3 User Roles and Characteristics

Role	Characteristics / Responsibilities
Passenger	Basic computer/mobile knowledge; searches trains, books tickets, views bookings and cancels eligible tickets.
Administrator	Authorized user responsible for maintaining train, route, schedule, fare and availability data.
Product Owner	Defines priorities, validates requirements and accepts completed features for the mini project.
Developer	Implements the application and resolves technical defects.
Tester	Designs and executes tests, verifies acceptance criteria and reports defects.

2.4 Operating Environment

- Client: modern web browser on desktop, laptop, tablet or mobile device.
- Application: web application running on a development/server environment.
- Database: relational database for users, trains, schedules, seats, bookings and cancellations.
- Network: local or internet-connected environment depending on deployment.

2.5 Constraints and Assumptions

- Only authenticated passengers can create or cancel their own bookings.
- The prototype assumes that train, schedule, fare and seat data are supplied by an authorized administrator.
- The system will not control physical railway operations or real-time train movement.
- Payment processing may be represented by a simulated/mock payment step for the academic prototype if a live payment service is not integrated.
- Cancellation/refund rules will follow the rules configured for the project and displayed to the passenger before confirmation.

3. External Interface Requirements

3.1 User Interface

- The system shall provide a clear home/search page for entering source, destination and journey date.
- Search results shall show train number/name, departure time, arrival time, duration, fare and seat availability.
- Booking screens shall clearly display selected train and passenger details before confirmation.
- The system shall provide confirmation and cancellation messages that clearly indicate success or failure.
- The interface shall provide readable labels, validation messages and navigation controls.

3.2 Hardware Interface

No dedicated railway hardware is required for the mini-project prototype. The application shall operate using standard computing/mobile devices capable of running a supported web browser.

3.3 Software Interfaces

- Frontend UI communicates with backend application services through HTTP/HTTPS APIs.
- Backend communicates with the relational database for persistent storage.
- Authentication and authorization are handled by the application backend.
- A mock/simulated payment interface may be used for the academic booking flow.

3.4 Communications

- Client-server communication shall use HTTP/HTTPS depending on deployment.
- API requests and responses shall use a structured format such as JSON.
- Authentication credentials and sensitive session information shall not be exposed in URLs.

4. System Features (Detailed Functional Requirements)

The following requirements use the format RTS-F-###. Each requirement is written as a testable 'shall' statement and is mapped to an acceptance criterion and test case reference.

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref	Dependencies / Comments
RTS-F-001	The system shall allow a new passenger to create an account using required personal and contact details.	Functional	High	Valid details create an account successfully. Test: TC-REG-01	Database; unique email/username
RTS-F-002	The system shall authenticate registered passengers using valid login credentials.	Functional	High	Valid credentials open the passenger session; invalid credentials are rejected. Test: TC-AUTH-01	User authentication module
RTS-F-003	The system shall allow passengers to search trains by source, destination and journey date.	Functional	High	Matching trains are displayed for valid search criteria. Test: TC-SEARCH-01	Train/schedule data
RTS-F-004	The system shall display train number/name, route, departure time, arrival time and fare for search results.	Functional	High	All required train details are visible for each result. Test: TC-SEARCH-02	Train schedule data
RTS-F-005	The system shall display the current seat availability for each available train/class.	Functional	High	Displayed availability matches stored availability. Test: TC-AVAIL-01	Seat inventory
RTS-F-006	The system shall allow an authenticated passenger to select a train and enter passenger details for a reservation.	Functional	High	Passenger can proceed only after required fields are valid. Test: TC-BOOK-01	Authentication; selected train
RTS-F-007	The system shall validate passenger details before creating a booking.	Functional	High	Invalid or missing mandatory details prevent booking. Test: TC-BOOK-02	Input validation
RTS-F-008	The system shall verify seat availability before confirming a booking.	Functional	High	Booking is rejected when no eligible seat is available. Test: TC-BOOK-03	Seat inventory
RTS-F-009	The system shall create a unique booking/PNR number for every successful reservation.	Functional	High	Every successful booking receives a unique reference. Test: TC-BOOK-04	Booking database
RTS-F-010	The system shall reduce the available seat count after a successful booking.	Functional	High	Seat availability decreases by the booked quantity. Test: TC-BOOK-05	Booking transaction
RTS-F-011	The system shall display the confirmed ticket containing booking reference, passenger, train, journey and fare details.	Functional	High	Confirmed ticket displays all mandatory details. Test: TC-TICKET-01	Successful booking
RTS-F-012	The system shall allow a passenger to view their active and previous bookings.	Functional	Medium	Only bookings belonging to the logged-in passenger are displayed. Test: TC-HISTORY-01	User identity; booking records
RTS-F-013	The system shall allow a passenger to cancel an eligible booking before the configured cancellation cutoff.	Functional	High	Eligible booking is cancelled and status changes to Cancelled. Test: TC-CANCEL-01	Cancellation rules
RTS-F-014	The system shall	Functional	High	Refund value follows	Fare and cancellation

	calculate and display the applicable refund amount when a booking is cancelled.			configured cancellation rules and is shown before confirmation. Test: TC-CANCEL-02	rules
RTS-F-015	The system shall restore the cancelled seat quantity to availability after a successful cancellation.	Functional	High	Cancelled seats are returned to the available inventory. Test: TC-CANCEL-03	Seat inventory
RTS-F-016	The system shall prevent a passenger from cancelling another passenger's booking.	Functional	High	Unauthorized cancellation attempt is rejected. Test: TC-SEC-03	Authorization
RTS-F-017	The system shall allow an administrator to add, update and deactivate train/schedule records.	Functional	High	Authorized admin changes are reflected in subsequent searches. Test: TC-ADMIN-01	Admin authorization
RTS-F-018	The system shall record booking and cancellation events with timestamps and relevant identifiers.	Functional	Medium	Events are stored and retrievable for audit/troubleshooting. Test: TC-AUDIT-01	Persistent logging

5. Non-Functional Requirements (Detailed)

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
RTS-NF-001	Search and normal booking operations shall return a user-visible response within 3 seconds for at least 90% of requests under normal project load.	Performance	High	90th percentile response time ≤ 3s. Test: TC-PERF-01
RTS-NF-002	The system shall maintain data consistency so that a seat cannot be successfully allocated to two concurrent bookings.	Reliability	High	Concurrent booking test shows no duplicate allocation. Test: TC-CONC-01
RTS-NF-003	The system shall provide clear validation and error messages without exposing internal database, stack-trace or credential information.	Usability/Security	High	Invalid operations show safe, understandable messages. Test: TC-ERR-01
RTS-NF-004	The system shall be available for normal project use and recover gracefully from temporary application/database failures.	Reliability	Medium	Service recovery procedure succeeds and data remains consistent. Test: TC-REC-01
RTS-NF-005	The system shall protect passwords using secure one-way password hashing rather than plaintext storage.	Security	High	Database inspection confirms passwords are not stored as plaintext. Test: TC-SEC-01
RTS-NF-006	The system shall enforce role-based access so that administrator functions are unavailable to ordinary passengers.	Security	High	Passenger account cannot access admin operations. Test: TC-SEC-02
RTS-NF-007	The user interface shall be responsive and usable on common desktop and mobile screen sizes.	Usability	Medium	Core booking flow remains usable on defined test viewport sizes. Test: TC-UX-01
RTS-NF-008	The system shall maintain an auditable record of booking and cancellation actions with timestamps.	Auditability	Medium	Audit records are generated for successful booking/cancellation actions. Test: TC-AUDIT-02

5.1 Security

5.1.1 Security Objectives

11. Protect passenger accounts and booking information from unauthorized access, modification or disclosure.

12. Preserve the integrity and consistency of reservation data so that bookings, cancellations and seat availability remain trustworthy.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance criteria / Test case ref
RTS-SR-001	The system shall store passenger passwords only as secure one-way hashes and shall never store plaintext passwords.	Security	High	Password storage review passes. Test: TC-SEC-01
RTS-SR-002	The system shall enforce authorization checks for passenger and administrator operations.	Security	High	Passenger cannot access admin endpoints or another passenger's booking. Test: TC-SEC-02/03
RTS-SR-003	The system shall use HTTPS/TLS for deployed client-server communication where supported by the deployment environment.	Security	High	Network/security configuration verifies encrypted transport. Test: TC-SEC-04
RTS-SR-004	The system shall validate and sanitize user-supplied input to reduce risks such as SQL injection and malicious script injection.	Security	High	Security input tests do not alter database structure or execute unintended scripts. Test: TC-SEC-05
RTS-SR-005	The system shall avoid placing passwords or other sensitive authentication data in application logs, URLs or error messages.	Security	High	Log and URL review contains no sensitive authentication data. Test: TC-SEC-06

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

- Correctness: bookings and cancellations must produce the expected ticket and seat-inventory state.
- Reliability: failed operations must not leave partial or inconsistent booking records.
- Security: authentication, authorization and input validation must protect user and reservation data.
- Usability: the passenger should be able to complete the search-to-booking flow with clear navigation and messages.
- Maintainability: the application should use modular components and clear naming so defects and enhancements can be handled efficiently.

6.2 Acceptance Criteria

- All High-priority functional requirements are implemented and verified.
- All mandatory NFR and security requirements pass their acceptance tests.
- A successful booking creates a unique booking reference and updates seat availability exactly once.
- A successful eligible cancellation changes the booking status and restores the correct seat quantity.
- Unauthorized users cannot access protected booking/admin operations.
- No critical defects remain open at final acceptance.

6.3 Acceptance Test Suites

Suite	Coverage
Authentication	Registration, login, invalid credentials and authorization
Train Search	Search criteria, train results, schedule and fare display
Availability	Seat availability and consistency
Booking	Passenger details, availability validation, confirmation and PNR
Cancellation	Eligibility, refund calculation, status update and seat restoration
Administration	Train/schedule management and access control
Security	Password handling, authorization, TLS, input validation and logging
Performance/UX	Response time and responsive interface

7. System Models and UML Use-Case Diagrams

The diagrams below show the major interactions between system actors and the Railway Ticket Reservation System. They are intentionally limited to the scope of the mini project.

7.1 UML Use-Case Diagram – Passenger

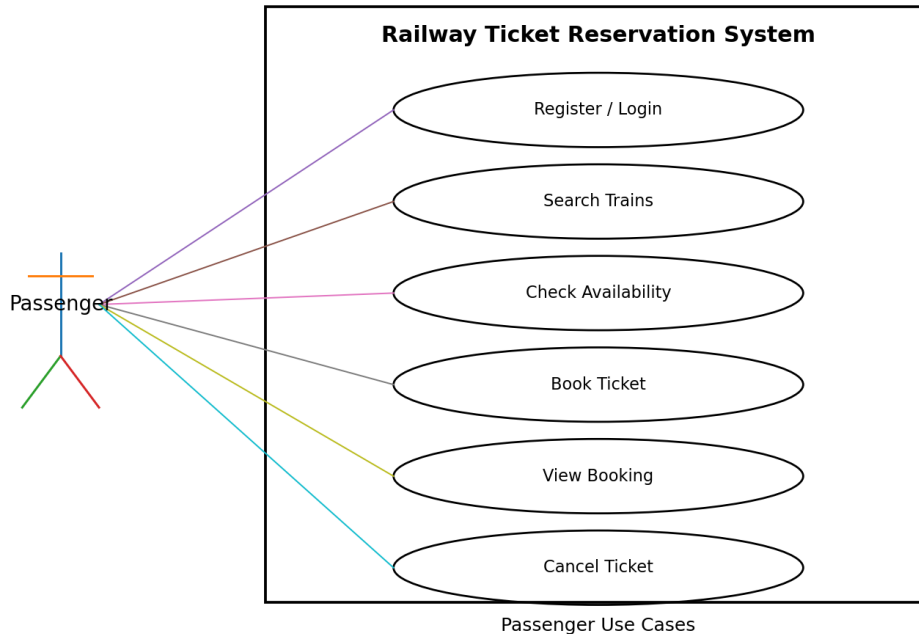


Figure 1: Passenger interactions with the Railway Ticket Reservation System.

7.2 UML Use-Case Diagram – Administrator

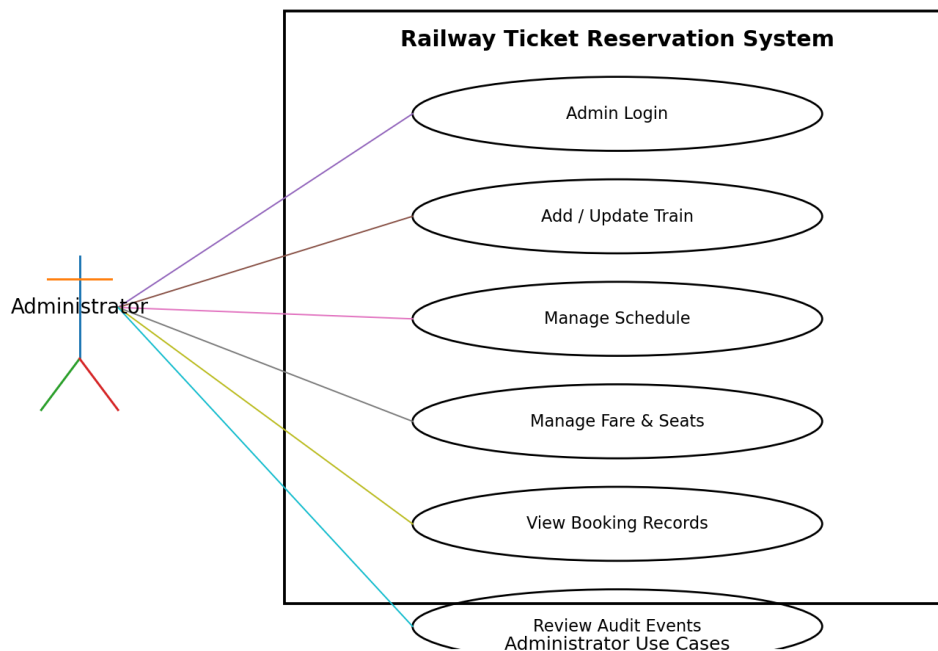


Figure 2: Administrator interactions with the Railway Ticket Reservation System.

8. Requirements Traceability Matrix (RTM)

The RTM links the requirements in this SRS to the planned design/module and test cases. Status is marked N (Not started) because implementation and testing are to be completed as part of the mini project.

Req ID	Requirement short	Section / Module	Test case(s)	Status
RTS-F-001	Registration	Authentication/User Module	TC-REG-01	N
RTS-F-002	Login	Authentication/User Module	TC-AUTH-01	N
RTS-F-003	Train search	Search Module	TC-SEARCH-01	N
RTS-F-004	Train details	Search Module	TC-SEARCH-02	N
RTS-F-005	Seat availability	Availability Module	TC-AVAIL-01	N
RTS-F-006	Passenger booking form	Booking Module	TC-BOOK-01	N
RTS-F-007	Passenger validation	Booking Module	TC-BOOK-02	N
RTS-F-008	Availability check	Booking Module	TC-BOOK-03	N
RTS-F-009	PNR creation	Booking Module	TC-BOOK-04	N
RTS-F-010	Seat decrement	Booking/Inventory Module	TC-BOOK-05	N
RTS-F-011	Ticket confirmation	Ticket Module	TC-TICKET-01	N
RTS-F-012	Booking history	Booking Module	TC-HISTORY-01	N
RTS-F-013	Cancellation	Cancellation Module	TC-CANCEL-01	N
RTS-F-014	Refund calculation	Cancellation Module	TC-CANCEL-02	N
RTS-F-015	Seat restoration	Cancellation/Inventory Module	TC-CANCEL-03	N
RTS-F-016	Ownership authorization	Authorization Module	TC-SEC-03	N
RTS-F-017	Admin management	Admin Module	TC-ADMIN-01	N
RTS-F-018	Audit events	Audit/Logging Module	TC-AUDIT-01	N
RTS-NF-001	Response time	Application/API	TC-PERF-01	N
RTS-NF-002	Concurrent seat consistency	Booking/Inventory Module	TC-CONC-01	N
RTS-NF-005	Password hashing	Authentication Module	TC-SEC-01	N
RTS-NF-006	Role-based access	Authorization Module	TC-SEC-02	N
RTS-SR-001	Secure password storage	Authentication Module	TC-SEC-01	N
RTS-SR-002	Authorization	Authorization Module	TC-SEC-02/03	N
RTS-SR-003	TLS	Deployment/API Layer	TC-SEC-04	N
RTS-SR-004	Input validation	Validation/API Layer	TC-SEC-05	N
RTS-SR-005	Sensitive-data logging	Logging Module	TC-SEC-06	N

Appendix A – Project Team and Responsibility Summary

Team Member	USN	Role	Primary SRS-related responsibility
Deeptanshu Kumar	PES1UG24AM357	Developer	Implementation feasibility and technical development
Amar	PES1UG24AM340	Tester	Acceptance criteria, test references and verification
Aryaprasad	PES1UG24AM331	Product Owner	Requirements, priorities and acceptance/sign-off

Appendix B – Assumptions for the Mini Project

- This SRS defines the academic mini-project scope and is not a specification for integration with a real railway reservation network.
- Train, schedule, fare and seat information are maintained within the project database or through project-defined APIs.
- The cancellation/refund policy is configurable and will be finalized by the team before implementation.
- The team may use simulated payment processing if live payment integration is outside the project scope.

End of Software Requirements Specification